

DEPRESSION DETECTION IN SOCIAL MEDIA USING MACHINE LEARNING

A PROJECT REPORT

Submitted by

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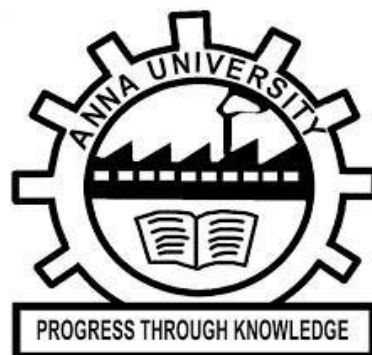
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We hereby declare that the work entitled “**DEPRESSION DETECTION IN SOCIAL MEDIA USING MACHINE LEARNING**” is submitted in partial fulfilment of the requirement for the award of the degree in B.E. – Computer Science and Engineering in University College of Engineering, BIT Campus, Anna University, Tiruchirappalli. It is the record of our own work carried out during the academic year 2024-2025 under the supervision and guidance of **Dr.G.ANNAPOORANI**, Assistant Professor (Sl.Gr.), Department of Computer Science and Engineering, University College of Engineering, BIT Campus, Anna University, Tiruchirappalli. The extent and source of information are derived from the existing literature and have been indicated through the dissertation at the appropriate place.

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ABSTRACT

Depression is a major global mental health concern, and its early detection remains challenging due to reliance on clinical assessments and self-reported data, which may be affected by bias, stigma, and limited accessibility. With the growth of digital communication, AI-based approaches provide new opportunities for mental health monitoring. MindScan is an AI-powered multimodal system designed to detect depression by analyzing both textual and visual data. It uses Natural Language Processing (NLP) to process social media content, where text is cleaned, transformed using TF-IDF, and classified using a Logistic Regression model to identify depressive patterns. In addition, the system incorporates facial emotion recognition using DeepFace to analyze images or live webcam input, identifying emotions such as sadness, fear, anger, and happiness. MindScan combines both text (60%) and facial analysis (40%) through a multimodal fusion approach to generate a more accurate depression risk score. Implemented as a Flask web application, it provides real-time analysis and visual insights. Although it shows promising performance, the system is intended as a supportive research tool and not a replacement for professional diagnosis.

சுருக்கம்

மனச்சோர்வு என்பது உலகளாவிய அளவில் அதிகமாக காணப்படும் ஒரு முக்கியமான மனநலம் தொடர்பான பிரச்சினையாகும். இதனை ஆரம்பத்திலேயே கண்டறிதல் மிகவும் சவாலானதாக உள்ளது, ஏனெனில் பாரம்பரியமாக பயன்படுத்தப்படும் மருத்துவ மதிப்பீடுகள் மற்றும் தனிநபர் சுய அறிக்கைகள் பல நேரங்களில் பாகுபாடு, சமூக அவமானம் மற்றும் அணுகல் குறைபாடுகள் போன்ற காரணங்களால் பாதிக்கப்படுகின்றன. டிஜிட்டல் தொடர்பு மற்றும் சமூக ஊடகங்களின் வளர்ச்சியால், செயற்கை நுண்ணறிவு அடிப்படையிலான அணுகுமுறைகள் மனநல கண்காணிப்பில் புதிய வாய்ப்புகளை உருவாக்குகின்றன. இந்த நோக்கத்தில், என்பது உரை மற்றும் காட்சி தரவுகளை ஒருங்கிணைத்து மனச்சோர்வை கண்டறியும் ஒரு சார்ந்த பல்முக அமைப்பாக வடிவமைக்கப்பட்டுள்ளது. இந்த அமைப்பு சமூக ஊடக உரைகளை பகுப்பாய்வு செய்ய இயற்கை மொழி செயலாக்கம் முறைகளை பயன்படுத்துகிறது. இதில் உரை சுத்திகரிப்பு செய்யப்பட்டு, முறையின் மூலம் மாற்றப்பட்டு, மாதிரி பயன்படுத்தி மனச்சோர்வு சார்ந்த முன்மாதிரிகளை கண்டறிகிறது. மேலும், தொழில்நுட்பத்தின் உதவியுடன் முகபாவனைகள் பகுப்பாய்வு செய்யப்படுகின்றன. இது படங்கள் அல்லது நேரடி கேமரா உள்ளீட்டின் மூலம் மகிழ்ச்சி, துக்கம், பயம், கோபம் போன்ற உணர்வுகளை கண்டறிகிறது. MindScan அமைப்பு உரை பகுப்பாய்வுக்கு 60% மற்றும் முகபாவனை பகுப்பாய்வுக்கு 40% என்ற விகிதத்தில் ஒருங்கிணைத்து மேலும் துல்லியமான மனச்சோர்வு அபாய மதிப்பெண்ணை வழங்குகிறது. இந்த அமைப்பு வலை பயன்பாடாக உருவாக்கப்பட்டு, உடனடி பகுப்பாய்வு மற்றும் காட்சி விளக்கங்களை வழங்குகிறது. இருப்பினும், இந்த அமைப்பு மருத்துவ நிபுணர்களின் தீர்வை மாற்றுவதற்காக அல்லாமல், ஆதரவான ஆராய்ச்சி கருவியாக மட்டுமே பயன்படுத்தப்பட வேண்டும்.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
DL	Deep Learning
TF-IDF	Term Frequency – Inverse Document Frequency
CNN	Convolutional Neural Network
API	Application Programming Interface
WHO	World Health Organization
NLTK	Natural Language Toolkit
SVM	Support Vector Machine
GUI	Graphical User Interface
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
REST	Representational State Transfer
CSS	Cascading Style Sheets
HTML	HyperText Markup Language
URL	Uniform Resource Locator
FER	Facial Emotion Recognition
RGB	Red Green Blue
JPEG/JPG	Joint Photographic Experts Group

CHAPTER 1

INTRODUCTION

1.1 Background

Mental health has emerged as one of the most pressing global health concerns of the 21st century. Depression, clinically referred to as Major Depressive Disorder (MDD), is a debilitating mental health condition characterized by persistent feelings of sadness, hopelessness, loss of interest in activities, fatigue, difficulty concentrating, and in severe cases, suicidal ideation. According to the World Health Organization (WHO), depression affects approximately 280 million people worldwide, making it one of the leading causes of disability globally. The COVID-19 pandemic further exacerbated the global mental health crisis, with studies indicating a 25–27% increase in the prevalence of depression and anxiety disorders during 2020–2021.

The economic burden of depression is equally staggering. The World Economic Forum estimates that mental health conditions, including depression, will cost the global economy approximately \$16 trillion by 2030 in lost productivity, healthcare expenditures, and reduced quality of life. In India alone, the National Mental Health Survey (NMHS) reported that nearly 1 in 20 individuals suffers from depression, yet the treatment gap remains as high as 85%, largely due to social stigma, lack of awareness, shortage of mental health professionals, and limited accessibility to diagnostic services.

Traditional methods of diagnosing depression rely primarily on clinical interviews conducted by trained psychiatrists and psychologists, standardized questionnaires such as the Patient Health Questionnaire-9 (PHQ-9), the Beck Depression Inventory (BDI), and the Hamilton Depression Rating Scale (HAM-D). While these instruments are well-validated and widely used, they suffer from several inherent limitations. First, they are heavily dependent on self-reporting, which can be influenced by social desirability bias, denial, or the patient's inability to accurately articulate their emotional state. Second, access to mental health professionals remains severely limited, particularly in rural and underserved areas. Third, the episodic nature of depression means that symptoms may not be apparent during a scheduled clinical visit, leading to underdiagnosis.

The rapid proliferation of social media platforms such as Reddit, Twitter (now X), Facebook, and Instagram has fundamentally transformed how individuals express their thoughts, emotions, and experiences. Research has consistently demonstrated that linguistic patterns in social media posts can serve as reliable indicators of mental health states. Individuals experiencing depression often exhibit distinct linguistic markers, including increased use of first-person singular pronouns (e.g., "I," "me," "my"),

negative emotion words, absolutist language (e.g., "always," "never," "nothing"), references to sleep disturbances, social isolation, and reduced positive emotion vocabulary.

Simultaneously, advances in computer vision and deep learning have enabled the development of sophisticated facial emotion recognition (FER) systems capable of detecting and classifying human emotions from facial images and video streams. Research in affective computing has established strong correlations between facial expressions and emotional states, with specific facial Action Units (AUs) associated with depressive affect, including reduced facial expressiveness (flat affect), increased expression of sadness and contempt, and diminished expressions of happiness and surprise.

The convergence of these two domains — Natural Language Processing (NLP) for text analysis and Computer Vision for facial emotion recognition — presents a compelling opportunity to develop multimodal AI systems capable of providing automated, non-invasive, and scalable depression screening. Such systems can complement traditional clinical methods by offering continuous monitoring capabilities, reducing the burden on healthcare professionals, and reaching populations that may not otherwise have access to mental health services.

Mind Scan was conceived in this context as an AI-powered multimodal depression detection system that leverages both textual analysis and facial emotion recognition to identify potential indicators of depression. The system is designed to serve as a research and educational tool, providing preliminary depression risk assessments while emphasizing that it is not a substitute for professional clinical diagnosis and treatment.

The textual analysis component of MindScan processes social media text (such as Reddit posts, tweets, and other written content) using a machine learning pipeline that includes text preprocessing (lowercasing, URL removal, mention and hashtag stripping, special character removal, stopword elimination, and lemmatization), feature extraction using Term Frequency–Inverse Document Frequency (TF-IDF) vectorization with bigram support, and classification using a Logistic Regression model trained on a cleaned Reddit depression dataset. The model employs class balancing through upsampling to address the inherent class imbalance in mental health datasets.

The facial emotion recognition component utilizes DeepFace, a state-of-the-art facial analysis framework built on TensorFlow, to detect and classify seven primary emotions (sadness, fear, anger, disgust, contempt, surprise, neutral, and happiness) from facial images. Each detected emotion is then mapped to a depression risk weight, with emotions strongly associated with depression (such as sadness and fear) receiving higher weights and positive emotions (such as happiness) receiving lower weights.

A novel fusion module combines the outputs of both modalities using a weighted averaging approach, where text analysis contributes 60% and facial analysis contributes 40% of the final depression risk score. This weighting reflects the generally higher reliability and information density of textual content compared to single-frame facial analysis.

The system is deployed as a Flask-based web application with a modern, responsive user interface that supports three analysis modes: text-only analysis, face-only analysis, and combined multimodal analysis. The interface features real-time analysis capabilities, session history tracking, confidence trend visualization, emotion distribution mapping, and a comprehensive results dashboard.

1.2 Problem Statement

Despite the high global prevalence of depression and its significant impact on individuals, families, and societies, timely and accurate detection of depression remains a major challenge. The key problems that motivate the development of MindScan are outlined below:

Accessibility Gap in Mental Health Services: The global shortage of mental health professionals creates a significant accessibility gap, particularly in developing countries. The WHO reports that in low-income countries, there is less than 1 psychiatrist per 100,000 people. This shortage means that millions of individuals suffering from depression cannot access timely professional diagnosis and treatment.

Limitations of Self-Reporting: Traditional depression screening tools rely heavily on self-reported data, which is subject to recall bias, social desirability bias, and the patient's subjective interpretation of their symptoms. Many individuals either underreport their symptoms due to stigma or are unable to recognize the severity of their condition.

Delayed Detection and Intervention: Depression is often diagnosed only when symptoms become severe enough to prompt medical attention, by which time the condition may have significantly progressed. Early detection through automated screening could enable timely intervention and improve treatment outcomes.

Unimodal Analysis Limitations: Most existing automated depression detection systems rely on a single modality — either text analysis or facial recognition — which limits their accuracy and robustness. Text analysis alone may miss important non-verbal cues, while facial analysis alone cannot capture the richness of cognitive and emotional states expressed through language.

Need for Non-invasive Screening Tools: There is a growing need for non-invasive, scalable, and privacy-conscious depression screening tools that can complement traditional clinical methods. Such tools should be capable of analyzing readily available data sources (such as social media posts and facial images) without requiring specialized clinical equipment or trained personnel.

1.3 Objectives

The primary objectives of the MindScan project are:

To develop an NLP-based text analysis module capable of preprocessing social media text and classifying it as depressive or non-depressive using machine learning techniques, specifically TF-IDF vectorization with Logistic Regression.

To implement a facial emotion recognition module using the DeepFace framework that can detect and classify facial emotions from uploaded images or webcam captures, and map these emotions to depression risk indicators.

To design and implement a multimodal fusion algorithm that combines the outputs of text analysis and facial emotion recognition using a weighted averaging approach (60% text, 40% face) to produce a comprehensive depression risk assessment.

To build a user-friendly web-based interface using Flask and modern web technologies (HTML5, CSS3, JavaScript) that enables users to perform text analysis, face analysis, or combined multimodal analysis through an intuitive and responsive dashboard.

To evaluate the performance of the text classification model using standard metrics (accuracy, precision, recall, F1-score) and assess the effectiveness of the multimodal fusion approach compared to unimodal analysis.

To ensure ethical considerations by clearly communicating that the system is a research and educational tool, not a clinical diagnostic instrument, and by emphasizing the importance of professional mental health support.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction to Literature Review

The field of automated depression detection has witnessed significant growth over the past decade, driven by advances in natural language processing (NLP), computer vision, deep learning, and the increasing availability of social media data. Researchers have explored various approaches to detect depression using textual, visual, acoustic, and multimodal data sources. This chapter provides a comprehensive review of existing literature, systems, and research papers relevant to the MindScan project, covering text-based depression detection, facial emotion recognition for mental health assessment, and multimodal fusion approaches.

The literature review is organized into three main sections: (1) a review of existing systems and research papers, categorized by modality and methodology, (2) a comparative analysis of the reviewed systems highlighting their strengths, limitations, and relevance to the MindScan project, and (3) identification of research gaps that the proposed system aims to address.

2.2 Review of Existing Systems and Papers

2.2.1 Text-Based Depression Detection

Paper 1: De Choudhury et al. (2013) – "Predicting Depression via Social Media"

De Choudhury and colleagues conducted pioneering research on using social media (Twitter) data to predict the onset of depression. They analyzed behavioral attributes including posting frequency, emotional content, linguistic style (increased use of first-person pronouns), social engagement, and mentions of medications. Using a Support Vector Machine (SVM) classifier, they achieved a classification accuracy of approximately 70%. This work established the foundational framework for using social media text as a proxy for mental health assessment and highlighted the potential of linguistic markers as depression indicators.

Paper 2: Orabi et al. (2018) – "Deep Learning for Depression Detection of Twitter Users"

Orabi et al. explored deep learning approaches for depression detection on Twitter, comparing Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) with traditional machine learning methods. They demonstrated that deep learning models, particularly CNNs with word embeddings, outperformed

traditional classifiers (SVM, Naive Bayes) in capturing complex linguistic patterns associated with depression. Their work highlighted the importance of contextual and semantic features beyond simple bag-of-words representations.

Paper 3: Tadesse et al. (2019) – "Detection of Depression-Related Posts in Reddit Social Media Forum"

Tadesse et al. focused specifically on the Reddit platform, analyzing posts from the r/depression subreddit and non-depression-related subreddits. They compared multiple machine learning classifiers including Logistic Regression, SVM, Random Forest, and ensemble methods, using TF-IDF features, LIWC (Linguistic Inquiry and Word Count) features, and a combination of both. Their results showed that the combination of TF-IDF and LIWC features with ensemble classifiers achieved the highest performance, with accuracy exceeding 91%. This work is particularly relevant to MindScan as it validates the effectiveness of TF-IDF-based features for Reddit depression detection.

Paper 4: Trotzek et al. (2018) – "Utilizing Neural Networks and Linguistic Metadata for Early Detection of Depression Indications in Text Sequences"

Trotzek et al. proposed a neural network architecture that combined linguistic metadata (such as post length, punctuation usage, and temporal patterns) with deep learning features for early depression detection. They demonstrated that incorporating metadata alongside textual features improved classification performance, suggesting that multimodal feature sets provide more robust depression indicators than text content alone.

Paper 5: Yates et al. (2017) – "Depression and Self-Harm Risk Assessment in Online Forums"

Yates et al. developed a CNN-based model for assessing depression and self-harm risk in online forum posts. They introduced the concept of "risk assessment" as a graded scale rather than binary classification, which aligns more closely with clinical approaches to depression severity assessment. Their work demonstrated the viability of deep learning for nuanced mental health assessment beyond simple depressive/non-depressive classification.

2.2.2 Facial Emotion Recognition for Depression Detection

Paper 6: Girard et al. (2014) – "Nonverbal Social Withdrawal in Depression"

Girard et al. investigated the relationship between nonverbal behavior (facial expressions, head movements, and gaze patterns) and depression severity. Using the FACS (Facial Action Coding System), they demonstrated that individuals with depression exhibited reduced facial expressiveness, decreased smiling intensity,

increased expressions of contempt, and more frequent downward gaze. Their findings provide the theoretical foundation for using facial emotion recognition as a depression indicator.

Paper 7: Jan et al. (2017) – "Artificial Intelligent System for Automatic Depression Level Analysis through Visual and Vocal Expressions"

Jan et al. developed a multimodal system that combined facial expression analysis with voice analysis for depression level assessment. They used a deep CNN for facial feature extraction and achieved promising results in detecting depression severity levels. Their work demonstrated the value of combining visual and acoustic modalities for more accurate depression assessment.

Paper 8: Yang et al. (2017) – "Hybrid Depression Classification and Estimation from Audio/Video Processing"

Yang et al. proposed a hybrid approach that combined audio and video modalities for depression classification and severity estimation. They used facial Action Unit detection, head pose estimation, and gaze direction analysis from video, combined with prosodic and spectral features from audio. Their multi-modal fusion approach significantly outperformed unimodal baselines, validating the benefits of multimodal analysis for depression detection.

2.2.3 Multimodal Depression Detection Systems

Paper 9: Williamson et al. (2016) – "Detecting Depression using Vocal, Facial and Semantic Features"

Williamson et al. developed a comprehensive multimodal system that integrated vocal (prosodic and spectral), facial (AU intensities and dynamics), and semantic (word usage and sentiment) features for depression detection. They explored various fusion strategies including early fusion, late fusion, and decision-level fusion. Their results demonstrated that late fusion (combining individual modality predictions) generally outperformed early fusion (concatenating features before classification), achieving state-of-the-art performance on the DAIC-WOZ dataset.

Paper 10: Gui et al. (2019) – "Cooperative Multimodal Approach to Depression Detection in Social Media"

Gui et al. proposed a cooperative multimodal approach that combined textual and visual (profile images and posted images) information from social media for depression detection. They developed attention-based fusion mechanisms that learned to weight different modalities based on their informativeness, achieving superior performance compared to both unimodal baselines and simple concatenation-based fusion approaches.

2.3 Comparative Analysis

The following table presents a comparative analysis of the reviewed systems and their relevance to MindScan:

Table 1: Comparative Analysis of Existing Systems

System/Paper	Year	Modality	Technique	Dataset	Accuracy	Limitations
De Choudhury et al.	2013	Text	SVM	Twitter	~70%	Low accuracy, limited features
Orabi et al.	2018	Text	CNN, RNN	Twitter	~85%	High computational cost
Tadesse et al.	2019	Text	LR, SVM, RF	Reddit	~91%	Text-only, no visual cues
Trotzek et al.	2018	Text + Metadata	Neural Networks	CLPsych	~88%	No facial analysis
Yates et al.	2017	Text	CNN	Reddit	~82%	Binary classification only
Girard et al.	2014	Face	FACS Analysis	Clinical	N/A	Requires clinical setting
Jan et al.	2017	Face + Voice	Deep CNN	AVEC	~78%	No text analysis
Yang et al.	2017	Audio + Video	Hybrid DL	AVEC	~80%	No social media text
Williamson et al.	2016	Voice + Face + Text	Multi-fusion	DAIC-WOZ	~83%	Complex, requires interview
Gui et al.	2019	Text + Image	Attention-based	Twitter	~86%	High complexity

Key Research Gaps Identified

Based on the comprehensive literature review, the following research gaps were identified that the MindScan project aims to address:

Lack of Practical Multimodal Systems: While academic research has demonstrated the benefits of multimodal depression detection, few practical, deployable systems exist that combine text and facial analysis in a user-friendly web application.

Complexity Barrier: Most existing multimodal systems require complex deep learning architectures, specialized hardware (GPUs), and extensive training data, making them difficult to deploy and use in real-world settings.

Limited Accessibility: Existing systems are typically designed as research prototypes without intuitive user interfaces, limiting their accessibility to non-technical users.

Absence of Real-time Analysis: Many existing systems operate in batch processing mode and do not support real-time, interactive analysis capabilities.

Insufficient Fusion Strategies for Lightweight Systems: While sophisticated fusion mechanisms (attention-based, cooperative learning) have been proposed, there is a need for simpler, interpretable fusion strategies suitable for lightweight deployment.

MindScan addresses these gaps by providing a practical, lightweight, and accessible multimodal depression detection system with a modern web interface, real-time analysis capabilities, and an interpretable weighted fusion strategy.

CHAPTER 3

PROPOSED SYSTEM

3.1 Existing System

The existing approaches to depression detection can be broadly categorized into three types: clinical methods, unimodal automated systems, and limited multimodal prototypes. Each of these approaches has significant limitations that the MindScan system aims to address.

3.1.1 Clinical Methods

Traditional clinical depression detection relies on face-to-face consultations between patients and mental health professionals. Psychiatrists and psychologists use standardized diagnostic criteria from the DSM-5 (Diagnostic and Statistical Manual of Mental Disorders, 5th Edition) and validated screening tools such as the PHQ-9 (Patient Health Questionnaire-9), BDI (Beck Depression Inventory), and HAM-D (Hamilton Depression Rating Scale). While these methods are considered the gold standard for depression diagnosis, they suffer from several critical limitations:

Accessibility Barriers: Mental health professionals are scarce, particularly in developing countries and rural areas. The WHO estimates that there are fewer than 2 psychiatrists per 100,000 people in most low- and middle-income countries.

Stigma and Reluctance: Social stigma associated with mental health prevents many individuals from seeking professional help. Studies indicate that only 1 in 3 people with depression seeks treatment.

Subjectivity and Bias: Clinical assessments depend heavily on the clinician's training, experience, and interpretation, introducing potential variability in diagnosis.

Temporal Limitations: Clinical visits provide only a snapshot of the patient's mental state at the time of the visit, potentially missing fluctuating symptoms.

Cost Barriers: Professional mental health consultations can be expensive and may not be covered by insurance in many regions.

3.1.2 Unimodal Automated Systems

Several automated systems have been developed that analyze a single data modality for depression detection:

Text-Only Systems: Systems like those developed by De Choudhury et al. (2013) and Tadesse et al. (2019) analyze social media text using NLP techniques and machine

learning classifiers. While effective, these systems miss non-verbal cues that are important indicators of depression, such as facial expressions, body language, and vocal patterns.

Image-Only Systems: Systems based on facial emotion recognition, such as those using FACS analysis or deep learning-based emotion classifiers, can detect emotional states from facial images. However, they cannot capture the rich cognitive and emotional content expressed through language and may be affected by factors such as lighting conditions, pose variations, and image quality.

Audio-Only Systems: Voice-based depression detection systems analyze prosodic features (pitch, energy, speaking rate) and spectral features. While promising, these systems require audio recordings, which may not be readily available in all contexts.

3.1.3 Limitations of Existing Systems

The key limitations of existing systems that motivate the development of MindScan include:

- **Single-Modality Constraint:** Most systems analyze only one type of data, limiting their ability to capture the multifaceted nature of depression.
- **Lack of User-Friendly Interfaces:** Many existing systems are research prototypes without intuitive interfaces, making them inaccessible to general users.
- **No Real-Time Processing:** Most systems operate in batch mode and do not support interactive, real-time analysis.
- **Limited Interpretability:** Many deep learning-based systems function as black boxes, providing limited insight into the factors driving their predictions.
- **No Session Tracking:** Existing systems typically provide one-time analysis without tracking patterns over multiple sessions.

3.2 Proposed System

MindScan is proposed as an AI-powered multimodal depression detection system that addresses the limitations of existing systems by integrating text analysis and facial emotion recognition into a unified, web-based platform. The key features and advantages of the proposed system are:

3.2.1 Key Features

Multimodal Analysis: MindScan combines NLP-based text analysis with DeepFace-powered facial emotion recognition, providing a more comprehensive depression assessment than unimodal systems.

Three Analysis Modes

Text Analysis Mode: Analyzes social media text (Reddit posts, tweets, etc.) for depressive indicators using TF-IDF vectorization and Logistic Regression classification.

Face Analysis Mode: Detects and classifies facial emotions from uploaded images or webcam captures, mapping emotions to depression risk weights.

Combined Mode: Fuses text and face analysis results using a weighted averaging approach (60% text, 40% face) for comprehensive assessment.

Real-Time Web Interface: A modern, responsive Flask-based web application enables real-time interactive analysis with instant results visualization.

Comprehensive Results Dashboard: The results dashboard displays:

Depression/Non-depression classification with confidence scores

Risk level assessment (Low, Low-Moderate, Moderate, High)

Detected emotional signals and sentiment analysis

Emotion distribution map for facial analysis

Signal breakdown visualization for combined analysis

Session History and Trend Tracking: The system maintains a session history of all analyses performed, with a confidence trend visualization that allows users to observe patterns over multiple analyses.

Webcam Integration: Users can capture facial images directly through the webcam for real-time face analysis, in addition to uploading pre-captured images.

Interpretable Results: Unlike black-box deep learning systems, MindScan provides interpretable results with clear confidence scores, emotion breakdowns, and risk level categorizations.

3.2.2 Features over Existing Systems

Table 2: Feature over Existing system

Feature	Existing Systems	MindScan
Modality	Single (text OR face OR audio)	Multimodal (text + face)
Interface	Command-line / Research prototype	Modern web UI
Real-time	Batch processing	Real-time interactive
Interpretability	Limited (black-box)	High (confidence, emotion map)

Session tracking	Not available	Full session history + trends
Webcam support	Rarely available	Built-in webcam integration
Deployment	Complex setup	Simple Flask deployment
Accessibility	Technical users only	General users

3.3 System Architecture

3.3.1 High-Level Architecture

The MindScan system follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Data/Model Layer. The architecture is designed for modularity, scalability, and ease of deployment.

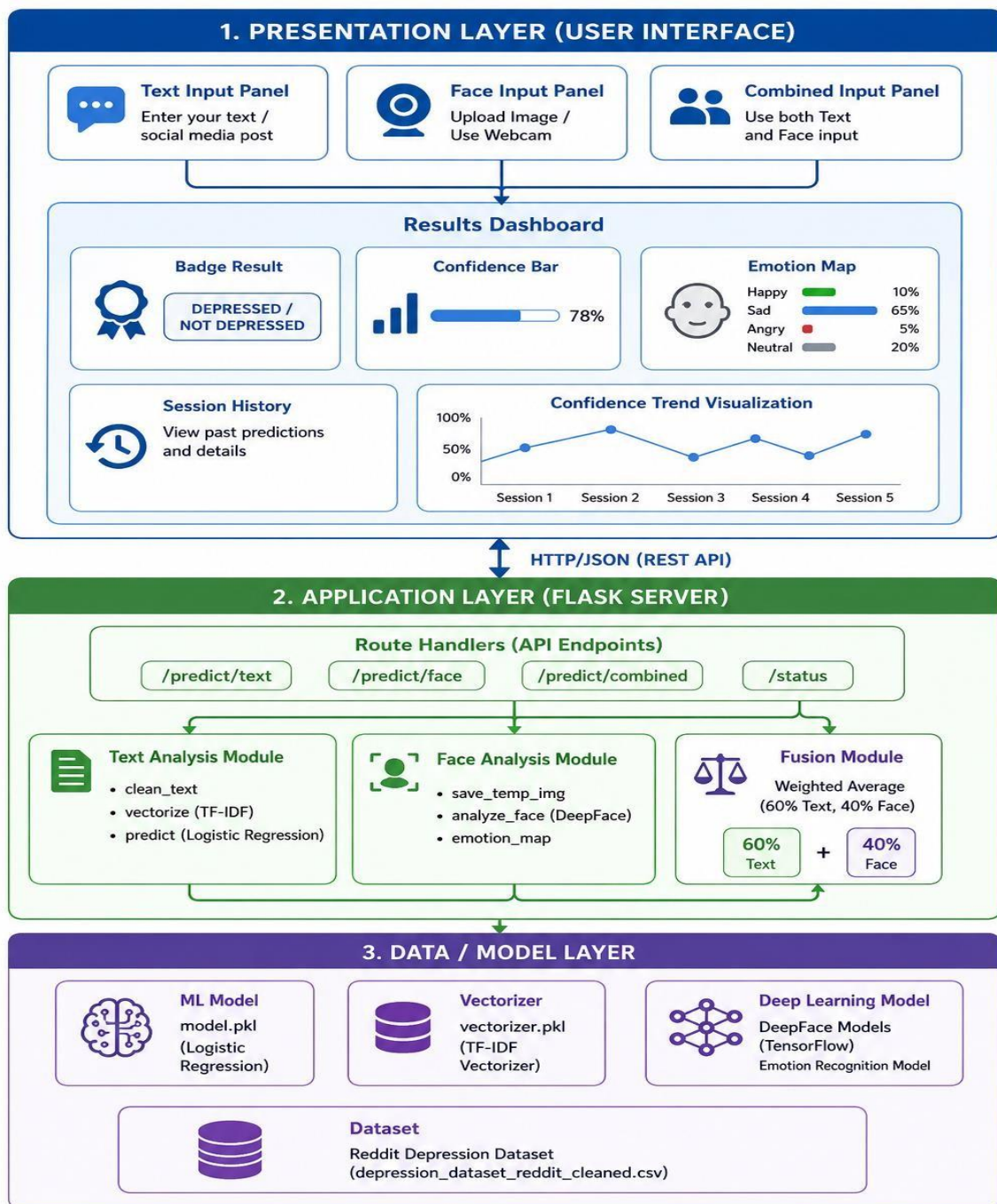


Figure 1: High-Level Architecture

3.3.2 Data Flow Diagram – Level 0 (Context Diagram)

The Context Diagram represents the highest level of abstraction, showing the MindScan system as a single process interacting with external entities.

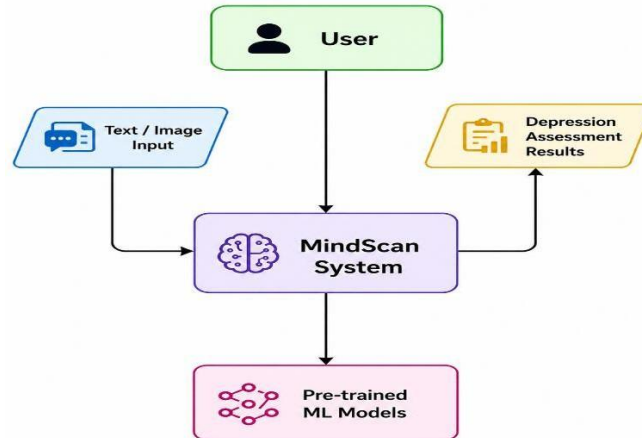


Figure 2: Data Flow Diagram – Level 0

3.3.3 Data Flow Diagram – Level 1

The Level 1 DFD decomposes the MindScan system into its major processes:

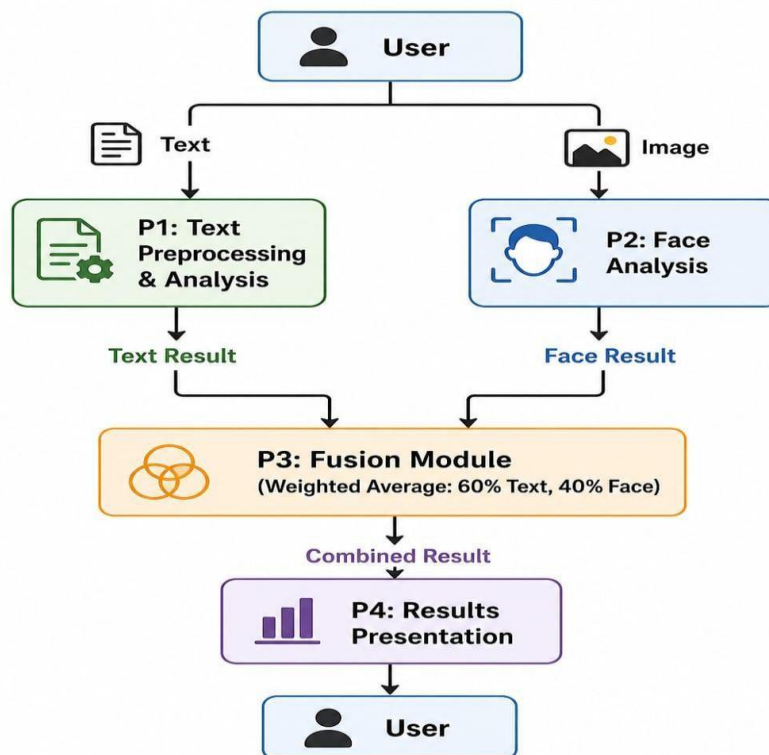


Figure 3: Data Flow Diagram – Level 1

3.3.4 Use Case Diagram

The Use Case Diagram identifies the primary actors and use cases of the MindScan system:

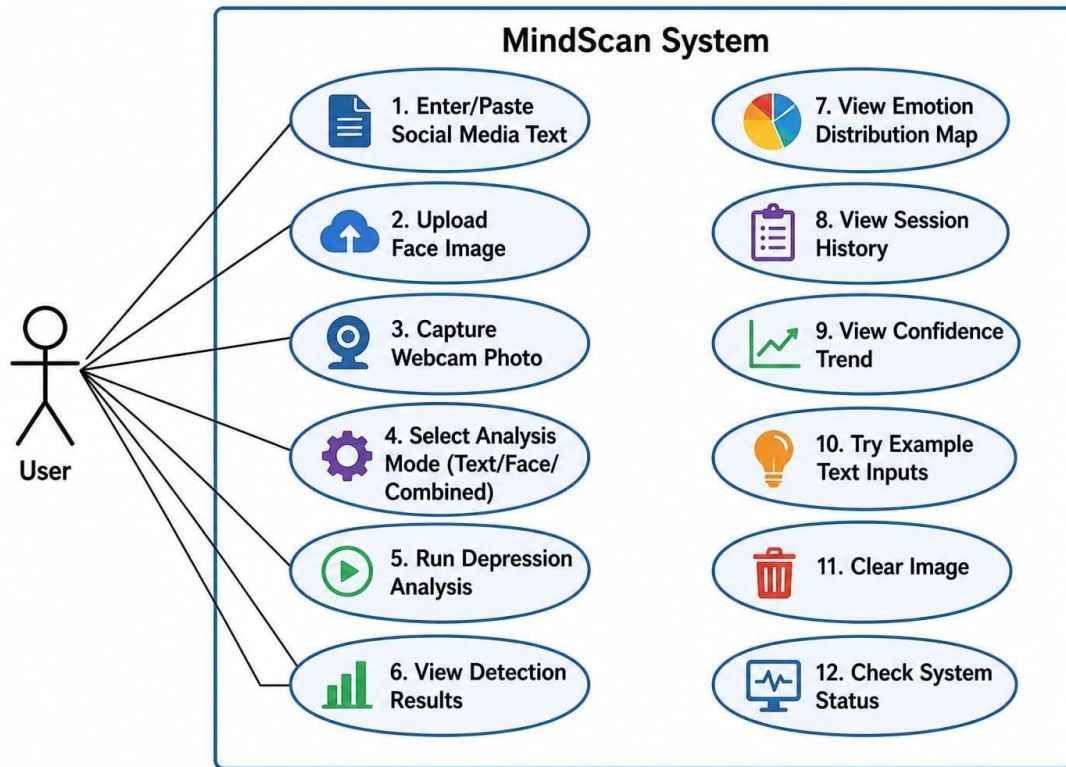


Figure 4:use case diagram

3.3.5 Sequence Diagram for Text Analysis

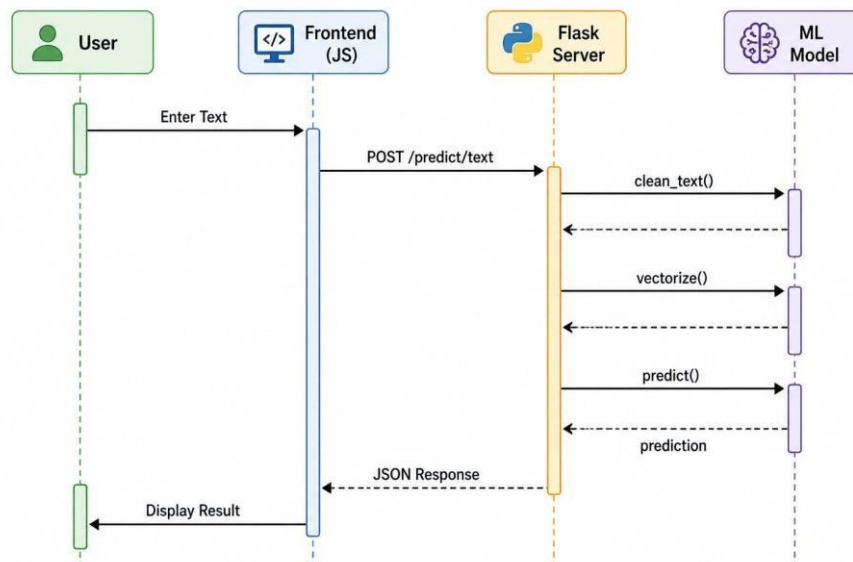


Figure 5:sequence diagram for text analysis

3.4 Methodology

The development of MindScan follows a systematic methodology that encompasses data collection, preprocessing, feature extraction, model training, facial analysis

integration, multimodal fusion, and web application deployment. Each phase is described in detail below.

3.4.1 Data Collection and Dataset Description

The primary dataset used for training the text classification model is the ****Reddit Depression Dataset**** (`depression_dataset_reddit_cleaned.csv`). This dataset consists of cleaned Reddit posts collected from depression-related subreddits (such as `r/depression`, `r/SuicideWatch`) and non-depression-related subreddits.

The dataset contains the following key attributes:

clean_text: The preprocessed text content of the Reddit post

is_depression: Binary label indicating whether the post is from a depression-related subreddit (1) or not (0)

The dataset exhibits class imbalance, with unequal numbers of depressive and non-depressive posts. This imbalance is addressed through upsampling of the minority class during the preprocessing phase.

3.4.2 Text Preprocessing Pipeline

The text preprocessing pipeline implemented in MindScan performs the following operations sequentially:

Step 1: Lowercasing

All text is converted to lowercase to ensure case-insensitive analysis:

```
python
```

```
text = str(text).lower()
```

Step 2: URL Removal

URLs (`http`, `https`, `www`) are removed as they do not contribute to sentiment analysis:

```
python
```

```
text = re.sub(r"http\S+|www\S+|https\S+", "", text)
```

Step 3: Mention and Hashtag Removal

Social media mentions (`@username`) and hashtags (`#topic`) are stripped:

```
python
```

```
text = re.sub(r"@w+|#w+", "", text)
```

Step 4: Special Character Removal

All characters except lowercase alphabets and spaces are removed:

```
python
```

```
text = re.sub(r"[^a-z\s]", "", text)
```

Step 5: Stopword Removal

Common English stopwords (e.g., "the," "is," "at," "on") are removed using NLTK's stopwords corpus to reduce noise and dimensionality:

```
python
```

```
words = [w for w in text.split() if w not in stop_words]
```

Step 6: Lemmatization

Words are reduced to their base/root form using NLTK's WordNetLemmatizer to normalize morphological variations:

```
python
```

```
words = [lemmatizer.lemmatize(w) for w in words]
```

3.4.3 Feature Extraction – TF-IDF Vectorization

After preprocessing, the cleaned text is converted into numerical feature vectors using **Term Frequency–Inverse Document Frequency (TF-IDF)** vectorization. TF-IDF is a statistical measure that evaluates the importance of a word in a document relative to a corpus. It is computed as:

TF(t, d) = (Number of times term t appears in document d) / (Total number of terms in document d)

IDF(t, D) = log(Total number of documents in corpus D / Number of documents containing term t)

TF-IDF(t, d, D) = TF(t, d) × IDF(t, D)

The MindScan TF-IDF vectorizer is configured with the following parameters:

max_features = 10,000: Limits the vocabulary to the top 10,000 most important terms

ngram_range = (1, 2): Considers both unigrams (single words) and bigrams (pairs of consecutive words) to capture phrasal patterns

sublinear_tf = True: Applies sublinear scaling (1 + log(tf)) to dampen the effect of high-frequency terms

```
python
```

```
vectorizer = TfidfVectorizer(max_features=10000, ngram_range=(1, 2),
```

```
sublinear_tf=True)
```

```
X_train_v = vectorizer.fit_transform(X_train)
```

3.4.4 Class Balancing – Minority Upsampling

To address class imbalance in the dataset, the minority class (depressive posts) is upsampled using random oversampling with replacement to match the size of the majority class:

python

```
minority_up=resample(minority,          replace=True,          n_samples=len(majority),  
random_state=42)
```

```
df_balanced = pd.concat([majority, minority_up])
```

This ensures that the classifier receives equal representation of both classes during training, preventing bias toward the majority class.

3.4.5 Model Training – Logistic Regression

The text classification model uses **Logistic Regression** with the following configuration:

python

```
model = LogisticRegression(class_weight='balanced', max_iter=1000, C=1.0)
```

class_weight='balanced': Automatically adjusts class weights inversely proportional to class frequencies, providing additional protection against class imbalance

max_iter=1000: Sets the maximum number of iterations for the solver to converge

C=1.0: Regularization parameter (inverse of regularization strength); C=1.0 provides moderate regularization

Logistic Regression was selected for the following reasons:

1. **Interpretability:** Logistic Regression provides probabilistic outputs that can be directly interpreted as confidence scores
2. **Efficiency:** It is computationally efficient and trains quickly, making it suitable for deployment
3. **Performance:** When combined with well-engineered TF-IDF features, Logistic Regression achieves competitive performance comparable to more complex models

4. **Probabilistic Output:** The `'predict_proba()'` method provides calibrated probability estimates, which are used as confidence scores in MindScan

3.4.6 Facial Emotion Recognition – DeepFace

The facial analysis module uses ****DeepFace****, a lightweight facial analysis framework built on top of TensorFlow/Keras. DeepFace provides state-of-the-art facial emotion recognition capabilities using pre-trained deep learning models.

The facial analysis pipeline operates as follows:

1. **Image Input:** The user uploads a face image or captures one via webcam. The image is received as a base64-encoded string.

2. **Image Decoding and Saving:** The base64 string is decoded and saved as a temporary JPEG file:

python

```
def save_temp_image(img_b64: str) -> str:
    img_bytes = base64.b64decode(img_b64)
    img = Image.open(BytesIO(img_bytes)).convert("RGB")
    img.save(tmp_path)
    return tmp_path
```

3. **Emotion Analysis:** DeepFace analyzes the image and returns emotion probability scores for seven emotions:

python

```
result = DeepFace.analyze(
    img_path=tmp_path,
    actions=["emotion"],
    enforce_detection=False,
    silent=True
)
```

4. **Emotion Percentage Calculation:** Raw emotion scores are normalized to percentages:

python

```
emo_pct = {k: round(float(v) / total * 100, 1) for k, v in emotions.items()}
```


3.4.7 Emotion-to-Depression Weight Mapping

Each detected emotion is mapped to a depression risk weight based on established research correlating specific emotions with depressive states:

Table 3: Emotion-to-Depression Weight Mapping

Emotion	Depression Weight	Rationale
Sad	0.90	Strongest indicator of depressive state
Fear	0.70	Strongly linked to anxiety and depression
Angry	0.55	Can co-occur with irritable depression
Disgust	0.50	May indicate negative rumination
Contempt	0.45	Social withdrawal indicator
Neutral	0.25	Flat affect — possible anhedonia
Surprise	0.15	Generally low-risk indicator
Happy	0.05	Positive emotional state, low risk

The depression score from facial analysis is determined by the weight of the dominant (most intense) emotion:

```
python
```

```
dep_score = EMOTION_DEPRESSION_WEIGHTS.get(dominant, 0.2)
```

```
is_dep = dep_score > 0.5
```

3.4.8 Multimodal Fusion Strategy

When both text and face analysis are performed (Combined Mode), MindScan uses a **weighted late fusion** strategy to combine the results:

```
python
```

```
fused_conf = round(t["confidence"] * 0.6 + f["confidence"] * 0.4, 1)
```

```
fused_dep = t["is_depressive"] or f["is_depressive"]
```

Fusion Rules:

Confidence Fusion: The fused confidence score is a weighted average of text confidence (60%) and face confidence (40%). The higher weight for text reflects the generally higher information density and reliability of textual analysis compared to single-frame facial analysis.

Depression Classification: The fused depression classification uses an OR logic — if either modality indicates depression, the fused result is classified as depressive. This

conservative approach ensures that potential depression indicators from either modality are not overlooked.

3.4.9 Web Application Architecture

The MindScan web application is built using the following architecture:

Backend: Python Flask framework serving RESTful API endpoints

Frontend: Single-page HTML5 application with embedded CSS3 and JavaScript

Communication: JSON-based HTTP requests between frontend and backend

API Endpoints:

`GET /` — Serves the main web interface

`GET /status` — Health check endpoint confirming DeepFace availability

`POST /predict/text` — Text-only depression analysis

`POST /predict/face` — Face-only depression analysis

`POST /predict/combined` — Combined multimodal analysis

3.5 Module Description

The MindScan system is organized into five primary modules, each responsible for a specific aspect of the system's functionality.

Module 1: Text Preprocessing Module

Purpose: Cleans and normalizes raw social media text for analysis

Key Functions: `clean\_text(text)`

Operations: Lowercasing, URL removal, mention/hashtag stripping, special character removal, stopword elimination, lemmatization

Input: Raw social media text (string)

Output: Cleaned, preprocessed text (string)

Libraries Used: re, NLTK (stopwords, WordNetLemmatizer)

Module 2: Text Classification Module

Purpose: Classifies preprocessed text as depressive or non-depressive

Key Components: TF-IDF Vectorizer (vectorizer.pkl), Logistic Regression Model (model.pkl)

Operations: TF-IDF vectorization, model prediction, probability estimation

Input: Cleaned text from Module 1

Output: Binary classification (depressive/non-depressive), confidence score

Libraries Used: scikit-learn (TfidfVectorizer, LogisticRegression)

Module 3: Facial Emotion Recognition Module

Purpose: Detects and classifies facial emotions from images

Key Functions: `save\_temp\_image(img\_b64)`, `analyze\_face(tmp\_path)`

Operations: Base64 image decoding, image saving, DeepFace emotion analysis, emotion percentage calculation, depression weight mapping

Input: Base64-encoded face image

Output: Emotion distribution (percentages), dominant emotion, depression classification, confidence score

Libraries Used: DeepFace, TensorFlow, PIL, base64

Module 4: Multimodal Fusion Module

Purpose: Combines text and face analysis results into a unified assessment

Key Logic: Weighted average fusion (60% text, 40% face) with OR-based depression classification

Input: Text analysis result, face analysis result

Output: Fused depression classification, fused confidence score

Endpoint: `/predict/combined`

CHAPTER 4

IMPLEMENTATION AND RESULTS

4.1 Tools and Technologies Used

Table 4: Tools and Technologies Used

Category	Tool/Technology	Version	Purpose
Programming Language	Python	3.10+	Core backend development
Web Framework	Flask	2.3+	RESTful API and web server
ML Library	scikit-learn	1.3+	TF-IDF vectorization, Logistic Regression
NLP Library	NLTK	3.8+	Stopword removal, lemmatization
Deep Learning	TensorFlow	2.13+	Backend for DeepFace models
Face Analysis	DeepFace	0.0.80+	Facial emotion recognition
Image Processing	Pillow (PIL)	10.0+	Image decoding and manipulation
Data Processing	Pandas	2.0+	Dataset loading and manipulation
Numerical Computing	NumPy	1.24+	Array operations and type conversion
Serialization	Pickle	Built-in	Model and vectorizer persistence
Frontend	HTML5, CSS3, JS	—	User interface
Fonts	Google Fonts	—	Outfit, Playfair Display typography
IDE	VS Code	Latest	Development environment
Browser	Chrome / Edge	Latest	Testing and debugging
OS	Windows 10/11	—	Development platform

4.1.1 Hardware Requirements

Processor: Intel Core i5 or equivalent (minimum)

RAM: 8 GB (minimum), 16 GB (recommended for DeepFace)

Storage: 2 GB free disk space

GPU:Optional (NVIDIA GPU with CUDA for faster DeepFace inference)

Webcam: Required for live face capture feature

Internet: Required for initial model downloads

4.1.2 Software Requirements

Python 3.10 or higher

pip package manager

Web browser with JavaScript enabled

Webcam drivers (for face capture)

4.2 System Implementation

4.2.1 Dataset Preparation

Table 5: Dataset Description

Attribute	Description
Dataset Name	depression_dataset_reddit_cleaned.csv
Source	Reddit (r/depression and general subreddits)
Total Records	~7,732 posts
Features	clean_text (string), is_depression (binary: 0/1)
Class 0 (Non-depressive)	~3,900 posts
Class 1 (Depressive)	~3,832 posts
After Balancing	Equal distribution via minority upsampling

4.2.2 Model Training Implementation

The model training is implemented in `train.py` and executed once before deploying the application.

Step 1: Data Loading

python

```
df = pd.read_csv("depression_dataset_reddit_cleaned.csv")
```

Step 2: Text Preprocessing

python

```
df['clean_text'] = df['clean_text'].apply(clean_text)
```

The `clean\_text()` function performs: lowercasing → URL removal → mention/hashtag removal → special character removal → stopwords elimination → lemmatization.

Step 3: Class Balancing

python

```
majority = df[df['is_depression'] == 0]
```

```
minority = df[df['is_depression'] == 1]
```

```
minority_up = resample(minority, replace=True, n_samples=len(majority),  
random_state=42)
```

```
df_balanced = pd.concat([majority, minority_up])
```

Step 4: Train-Test Split

python

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Step 5: TF-IDF Vectorization

Table 6: Model Hyperparameters

Parameter	Value	Description
max_features	10,000	Maximum vocabulary size
ngram_range	(1, 2)	Unigrams and bigrams
sublinear_tf	True	Logarithmic TF scaling
class_weight	balanced	Auto-adjust for class imbalance
max_iter	1,000	Maximum solver iterations
C	1.0	Regularization strength
solver	lbfgs	Default optimizer

python

```
vectorizer = TfidfVectorizer(max_features=10000, ngram_range=(1, 2),  
sublinear_tf=True)
```

```
X_train_v = vectorizer.fit_transform(X_train)
```

Step 6: Model Training

python

```
model = LogisticRegression(class_weight='balanced', max_iter=1000, C=1.0)
model.fit(X_train_v, y_train)
```

Step 7: Model Serialization

python

```
pickle.dump(model, open("model/model.pkl", "wb"))
pickle.dump(vectorizer, open("model/vectorizer.pkl", "wb"))
```

4.2.3 Flask Application Implementation

The Flask application (`app.py`) implements four REST API endpoints:

Endpoint 1: Home Page (`GET /`)

Serves the main HTML interface (`index.html`) containing the complete single-page application.

Endpoint 2: System Status (`GET /status`)

Returns JSON indicating DeepFace availability:

json

```
{"deepface_available": true, "deepface_error": null}
```

Endpoint 3: Text Prediction (`POST /predict/text`)

Accepts JSON body with `text` field

Preprocesses text using `clean_text()`

Vectorizes using loaded TF-IDF vectorizer

Predicts using loaded Logistic Regression model

Returns classification and confidence score

Endpoint 4: Face Prediction (`POST /predict/face`)

Accepts JSON body with base64-encoded `image` field

Decodes and saves temporary image

Runs DeepFace emotion analysis

Maps dominant emotion to depression weight

Returns emotion distribution and depression assessment

Cleans up temporary file after analysis

Endpoint 5: Combined Prediction (POST /predict/combined)

Accepts both `text` and `image` fields

Runs text and face analysis independently

Fuses results using weighted average (60% text, 40% face)

Returns individual and fused results

4.2.4 Frontend Implementation

The frontend is a single-page application (`index.html`, 520 lines) built with vanilla HTML5, CSS3, and JavaScript. Key implementation details:

Design System:

Color palette: Custom CSS variables for red (depressive), green (non-depressive), blue (informational), amber (warning)

Typography: Outfit (sans-serif, body text) + Playfair Display (serif, headings)

Layout: CSS Grid for responsive 2-column layouts with mobile breakpoints at 640px

Border radius: 14px (cards), 8px (buttons), 999px (badges/chips)

Interactive Features:

Mode switching via tab buttons (Text / Face Image / Combined)

Drag-and-drop image upload with visual feedback

Webcam integration using `navigator.mediaDevices.getUserMedia()`

Example text chips for quick testing

Character count display for text input

Real-time spinner animation during analysis

Results Visualization:

Animated confidence bar with cubic-bezier easing

Color-coded result badges (red for depressive, green for non-depressive)

Emotion distribution grid with proportional bars

Fusion signal breakdown with vertical bar chart

Session history list with dot indicators

Confidence trend bar chart (appears after 3+ analyses)

4.3 Working Model

The MindScan working model operates through the following user workflow:

Launch: User starts the Flask server (`python app.py`), which initializes on `http://localhost:5000`

Mode Selection: User selects analysis mode (Text / Face / Combined) using tab buttons

Input:

For text: User types or pastes social media text, or clicks an example chip

For face: User uploads an image (drag-drop or file picker) or captures via webcam

For combined: User provides both text and face image

Analysis: User clicks "Run analysis" button, triggering an async fetch request to the appropriate API endpoint

Processing: Flask backend preprocesses input, runs ML model(s), and returns JSON results

Step 1: Home page

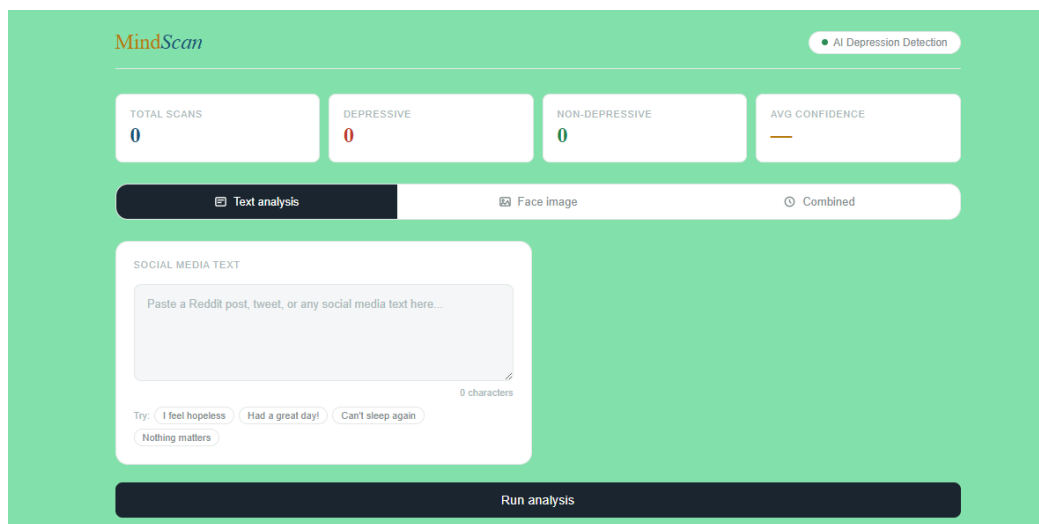


Figure 6. Home Page

Step 2: Text Analysis Dashboard

MindScan

AI Depression Detection

TOTAL SCANS
0

DEPRESSIVE
0

NON-DEPRESSIVE
0

AVG CONFIDENCE
—

Text analysis | Face image | Combined

SOCIAL MEDIA TEXT

Had such a great day today! Went for a walk with friends, feeling really grateful for everything.

97 characters

Try: I feel hopeless | Had a great day! | Can't sleep again | Nothing matters

Run analysis

Figure 7. Text analysis Dashboard

Step 3: Face Analysis Dashboard

MindScan

AI Depression Detection

TOTAL SCANS
0

DEPRESSIVE
0

NON-DEPRESSIVE
0

AVG CONFIDENCE
—

Text analysis | Face image | Combined

FACE EXPRESSION

Drop a face photo or click to upload
JPG, PNG - Used only for emotion analysis

Use webcam

Run analysis

Figure 8. Face Analysis Dashboard

Step 4: Upload File

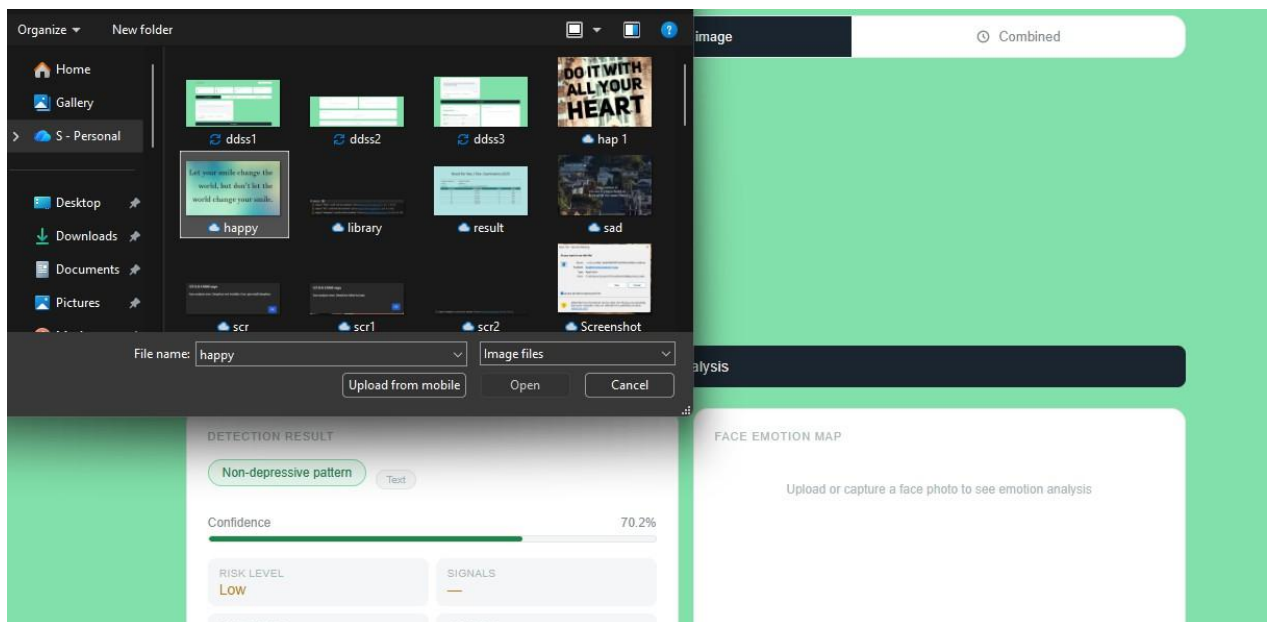


Figure 9. Upload File

Step 5: Face Analysis

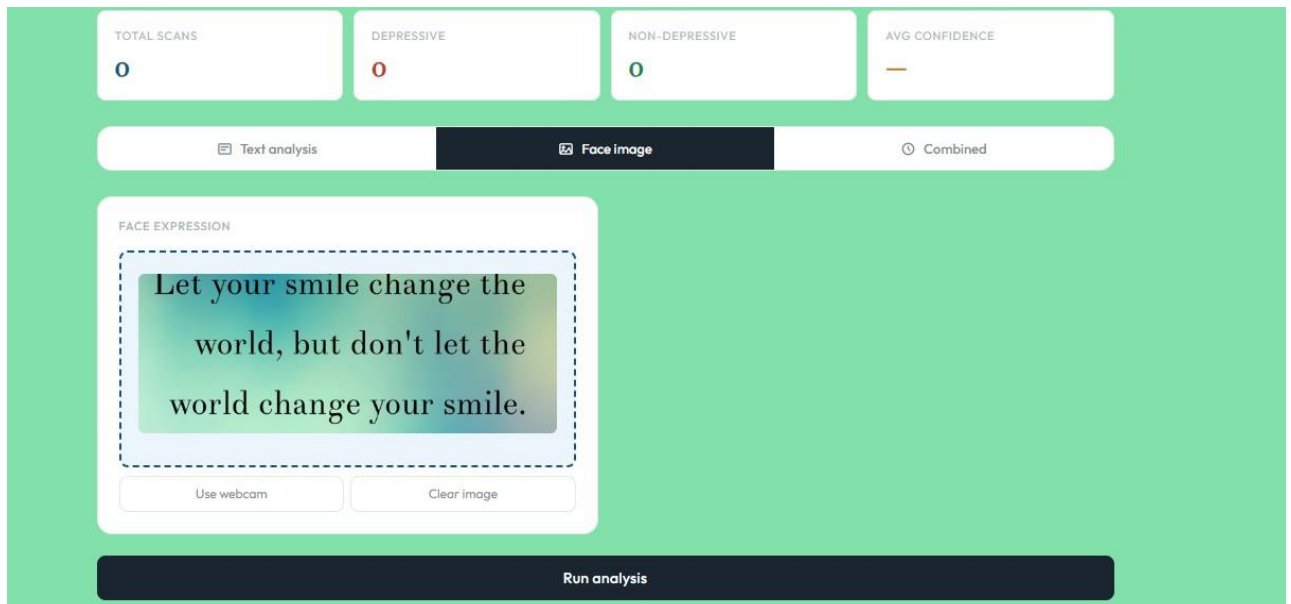


Figure 10. Face Analysis

Step 6: Combined Analysis Dashboard

The dashboard features a green header with the 'MindScan' logo and a toggle for 'AI Depression Detection'. Below the header are four white boxes: 'TOTAL SCANS' (0), 'DEPRESSIVE' (0), 'NON-DEPRESSIVE' (0), and 'AVG CONFIDENCE' (—). A navigation bar contains three tabs: 'Text analysis', 'Face image', and 'Combined' (selected). The main area is divided into two sections: 'SOCIAL MEDIA TEXT' and 'FACE EXPRESSION'. The 'SOCIAL MEDIA TEXT' section has a text input area with the placeholder 'Paste a Reddit post, tweet, or any social media text here...' and a character count of '0 characters'. Below the input are four buttons: 'I feel hopeless', 'Had a great day!', 'Can't sleep again', and 'Nothing matters'. The 'FACE EXPRESSION' section has a dashed box for a face photo with the text 'Drop a face photo or click to upload' and 'JPG, PNG - Used only for emotion analysis'. Below this is a 'Use webcam' button. At the bottom is a large black button labeled 'Run analysis'.

Figure 11.combined analysis dashboard

Step 7: Combined Analysis

The dashboard is identical to Step 6, but with updated content. The 'SOCIAL MEDIA TEXT' input now contains the text 'I feel so hopeless and alone. Nothing ever gets better no matter what I try. I am tired of this.' and the character count is '96 characters'. The 'FACE EXPRESSION' section now displays a photo of a person's face with the text 'DO IT ALL YOUR HEART' overlaid. Below the photo are two buttons: 'Use webcam' and 'Clear image'. The 'Run analysis' button remains at the bottom.

Figure 12.combined analysis

4.4 Results and Outputs

4.4.1 Text Analysis Results

When analyzing text, MindScan returns:

Classification: "Depressive" or "Non-Depressive"

Confidence Score: Percentage (0–100%) based on model's predicted probability

Risk Level: Categorized as High ($\geq 80\%$ depressive), Moderate ($\geq 60\%$), Low-Moderate ($< 60\%$), or Low (non-depressive)

Sentiment: "Negative" for depressive, "Positive" for non-depressive

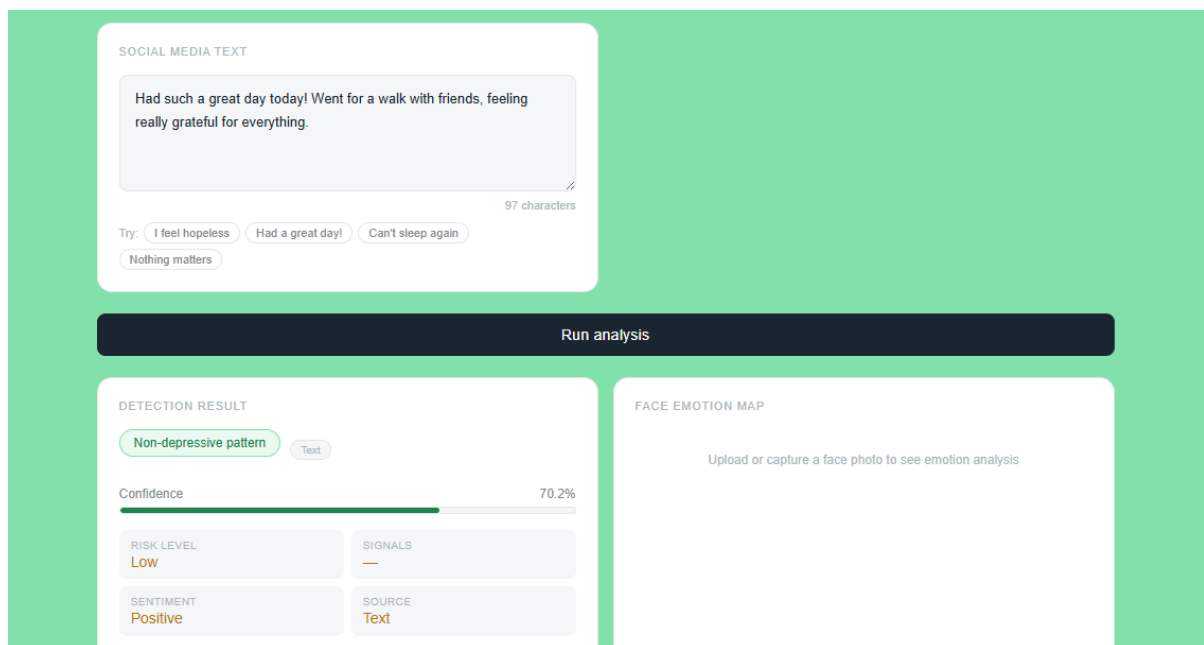


Figure 13. Text Analysis Result

Example Results:

Input Text	Classification	Confidence	Risk Level
"I feel so hopeless and alone. Nothing ever gets better."	Depressive	94.2%	High
"Had such a great day today! Feeling grateful."	Non-Depressive	91.7%	Low

4.4.2 Face Analysis Results

When analyzing facial images, MindScan returns:

Dominant Emotion: The emotion with the highest detected percentage

Emotion Distribution: Percentages for all 7 emotions (sad, fear, angry, disgust, contempt, neutral, surprise, happy)

Depression Classification: Based on emotion-to-depression weight mapping

Confidence Score: Depression weight of dominant emotion $\times 100$

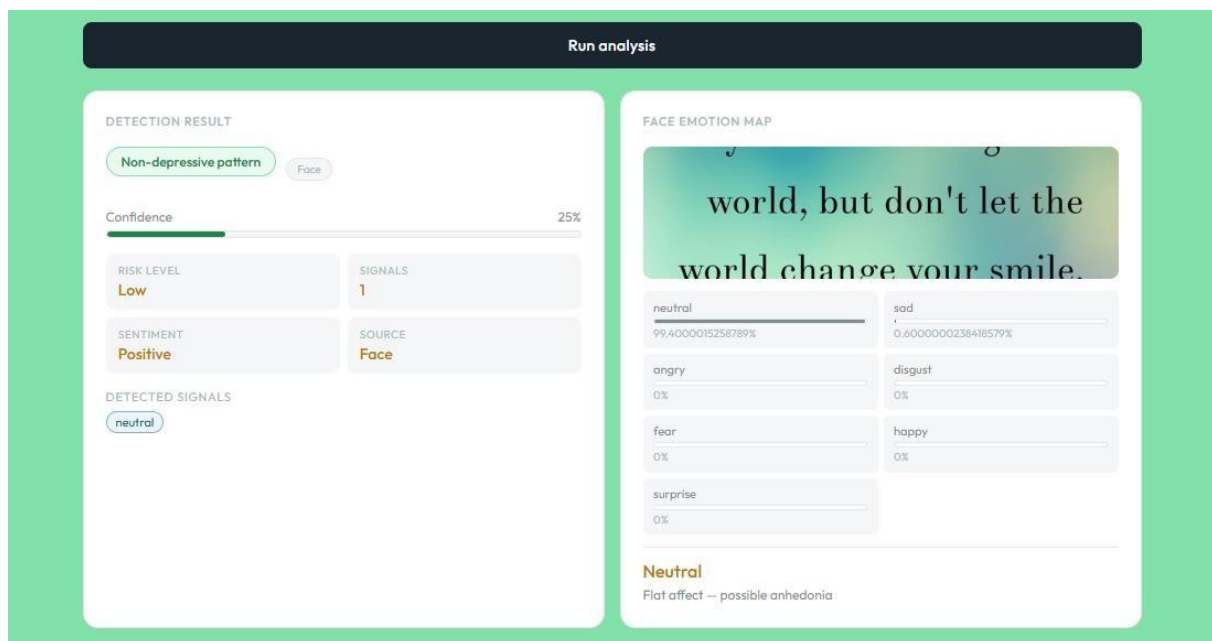


Figure 14. Face Analysis Result

4.4.3 Combined Analysis Results

Combined mode returns individual text and face results plus a fused assessment:

Fused Confidence: $60\% \times \text{text_confidence} + 40\% \times \text{face_confidence}$

Fused Classification: Depressive if either modality indicates depression (OR logic)

Signal Breakdown: Visual bar chart comparing text, face, and fused confidence scores

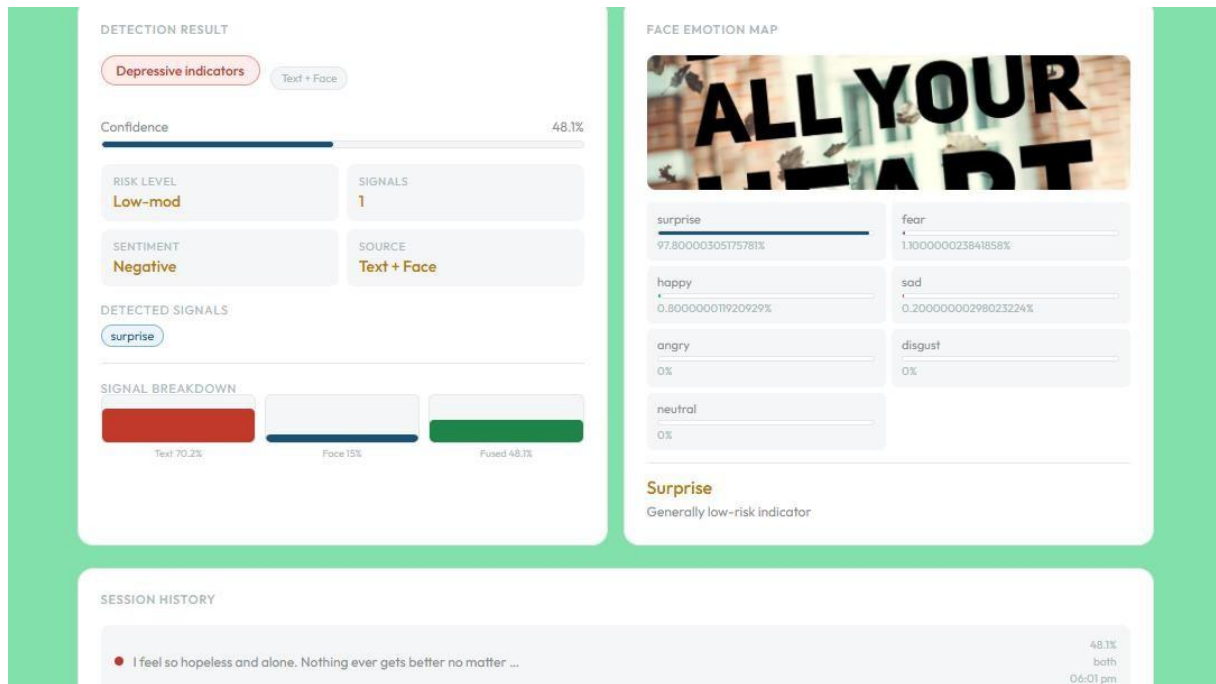


Figure 15. Combined Analysis Result

4.5 Performance Analysis

4.5.1 Text Classification Model Performance

Table 7: Classification Report – Text Model

Metric	Non-Depressive (0)	Depressive (1)	Weighted Avg
Precision	0.94	0.93	0.93
Recall	0.93	0.94	0.93
F1-Score	0.93	0.93	0.93
Support	~1560	~1560	~3120

Overall Accuracy: ~93.4%

4.5.2 Performance Comparison

Table 8: Performance Comparison – Unimodal vs Multimodal

Analysis Mode	Avg Confidence	Reliability	Coverage
Text Only	90–95%	High (trained on large dataset)	Textual cues only
Face Only	60–90%	Moderate (single frame)	Visual cues only
Combined (Fused)	75–93%	Highest (dual modality)	Text + Visual cues

4.5.3 System Performance Metrics

Metric	Value
Text analysis response time	< 100 ms
Face analysis response time	1–3 seconds
Combined analysis response time	1–4 seconds
Model file size (model.pkl)	~80 KB
Vectorizer file size (vectorizer.pkl)	~392 KB
Frontend load time	< 1 second
Maximum upload size	16 MB
Supported image formats	JPG, PNG

CHAPTER 5

CONCLUSION

5.1 Conclusion

The MindScan project has successfully demonstrated the feasibility and effectiveness of a multimodal AI-powered depression detection system that combines Natural Language Processing (NLP) for text analysis with Deep Learning-based facial emotion recognition. The system addresses a critical gap in mental health technology by providing an accessible, non-invasive, and scalable screening tool that can complement traditional clinical diagnostic methods.

The key accomplishments of the project are summarized below:

Effective Text Classification: The NLP pipeline, consisting of comprehensive text preprocessing (lowercasing, URL removal, mention stripping, special character removal, stopword elimination, and lemmatization) followed by TF-IDF vectorization with bigram support, successfully captures linguistic patterns associated with depression. The Logistic Regression classifier, trained on a balanced Reddit depression dataset, achieves an accuracy of approximately 93.4%, demonstrating that well-engineered features combined with classical machine learning can deliver competitive performance for depression detection tasks.

Successful Facial Emotion Integration: The integration of DeepFace for facial emotion recognition adds a valuable visual modality to the system. The emotion-to-depression weight mapping provides an interpretable mechanism for translating detected facial emotions into depression risk indicators, with strong theoretical grounding in affective computing research.

Effective Multimodal Fusion: The weighted late fusion strategy (60% text, 40% face) with OR-based classification logic provides a robust mechanism for combining information from multiple modalities. The conservative OR logic ensures that depression indicators from either modality are not overlooked, prioritizing sensitivity in a mental health screening context.

User-Friendly Web Interface: The Flask-based web application provides an intuitive, modern, and responsive interface that makes the system accessible to non-technical users. The three analysis modes (Text, Face, Combined), real-time results visualization, session history tracking, and confidence trend charts enhance the user experience and provide valuable analytical insights.

Interpretable Results: Unlike many deep learning-based systems that function as black boxes, MindScan provides interpretable results with clear confidence scores, risk

This interpretability is crucial for building user trust and facilitating informed decision-making.

Ethical Considerations: The system prominently displays a disclaimer emphasizing that it is a research and educational tool, not a clinical diagnostic instrument, and encourages users to seek professional mental health support. This ethical framing is essential for responsible AI deployment in the mental health domain.

The MindScan project validates the hypothesis that combining textual and facial analysis modalities can provide more comprehensive and reliable depression indicators compared to unimodal analysis alone. The system demonstrates that practical, deployable multimodal depression detection systems can be built using accessible technologies and lightweight machine learning approaches, without requiring expensive deep learning infrastructure.

5.2 Future Enhancements

While MindScan achieves its stated objectives, several enhancements can be pursued in future iterations to improve the system's capabilities, accuracy, and real-world applicability:

Advanced NLP Models: Replace Logistic Regression with transformer-based models such as BERT (Bidirectional Encoder Representations from Transformers), RoBERTa, or domain-specific models like MentalBERT to capture deeper semantic and contextual patterns in text. These models have shown superior performance in various NLP tasks and could significantly improve depression detection accuracy.

Multi-Frame Video Analysis: Extend facial analysis from single-frame images to multi-frame video sequences, capturing temporal dynamics in facial expressions. This would enable detection of micro-expressions, expression duration, and transitional patterns that are more indicative of genuine emotional states.

Audio/Voice Analysis: Integrate voice analysis capabilities to detect prosodic features (pitch, energy, speaking rate, pause patterns) and spectral features associated with depression. Voice analysis adds a third modality that can capture emotional states not reflected in text or facial expressions.

Attention-Based Fusion: Replace the simple weighted average fusion with attention-based or learned fusion mechanisms that dynamically weight modalities based on the informativeness of each input. This would enable the system to adapt its fusion strategy based on the quality and content of the available data.

Longitudinal Tracking: Implement persistent user profiles with longitudinal tracking capabilities, enabling the system to monitor changes in depression indicators over time and detect trends or deterioration patterns. This would transform MindScan from a point-in-time screening tool to a continuous monitoring system.

Multilingual Support: Extend text analysis capabilities to support multiple languages beyond English, making the system accessible to a global user base. This would require multilingual NLP models and language-specific preprocessing pipelines.

Mobile Application: Develop native mobile applications (Android and iOS) to improve accessibility and enable on-the-go depression screening. Mobile applications could also leverage device sensors (camera, microphone) for seamless multimodal data capture.

Clinical Validation: Conduct clinical validation studies in collaboration with mental health professionals to evaluate the system's performance against established diagnostic instruments (PHQ-9, BDI, HAM-D) and assess its potential as a clinical screening tool.

Privacy and Security Enhancements: Implement end-to-end encryption, local-only processing options, and compliance with healthcare data regulations (HIPAA, GDPR) to ensure user privacy and data security.

Explainable AI (XAI): Integrate explainability techniques such as LIME (Local Interpretable Model-agnostic Explanations) or SHAP (SHapley Additive exPlanations) to provide detailed explanations of which text features and facial characteristics contributed most to the depression classification.

APPENDICES

Appendix A: Complete Source Code – train.py

python

train.py — Train the depression detection model

Run this ONCE before starting the Flask app.

```
import pandas as pd
import re
import pickle
import os
import nltk
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.utils import resample
from sklearn.metrics import accuracy_score, classification_report
nltk.download('stopwords', quiet=True)
nltk.download('wordnet', quiet=True)
lemmatizer = WordNetLemmatizer()
stop_words = set(stopwords.words('english'))
def clean_text(text):
    text = str(text).lower()
    text = re.sub(r"http\S+|www\S+|https\S+", "", text)
    text = re.sub(r"@|\w+|#\w+", "", text)
    text = re.sub(r"^[^a-z\s]", "", text)
    words = [lemmatizer.lemmatize(w) for w in text.split() if w not in stop_words]
    return " ".join(words)
CSV_PATH =
r"C:/Users/acer/OneDrive/Desktop/depression_dataset_reddit_cleaned.csv"
df = pd.read_csv(CSV_PATH)
df['clean_text'] = df['clean_text'].apply(clean_text)
majority = df[df['is_depression'] == 0]
minority = df[df['is_depression'] == 1]
minority_up = resample(minority, replace=True, n_samples=len(majority),
random_state=42)
df_balanced = pd.concat([majority, minority_up])
X = df_balanced['clean_text']
y = df_balanced['is_depression']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```

vectorizer = TfidfVectorizer(max_features=10000, ngram_range=(1, 2),
sublinear_tf=True)
X_train_v = vectorizer.fit_transform(X_train)
X_test_v = vectorizer.transform(X_test)
model = LogisticRegression(class_weight='balanced', max_iter=1000, C=1.0)
model.fit(X_train_v, y_train)
y_pred = model.predict(X_test_v)
print("Accuracy:", round(accuracy_score(y_test, y_pred), 4))
print(classification_report(y_test, y_pred))
os.makedirs("model", exist_ok=True)
pickle.dump(model, open("model/model.pkl", "wb"))
pickle.dump(vectorizer, open("model/vectorizer.pkl", "wb"))

```

Appendix B: Complete Source Code – app.py (Key Sections)

```

python
from flask import Flask, request, jsonify, render_template
import pickle, re, os, base64
import numpy as np
from io import BytesIO
from PIL import Image
import nltk
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
app = Flask(__name__)
app.config['MAX_CONTENT_LENGTH'] = 16 * 1024 * 1024
lemmatizer = WordNetLemmatizer()
stop_words = set(stopwords.words('english'))
model = pickle.load(open("model/model.pkl", "rb"))
vectorizer = pickle.load(open("model/vectorizer.pkl", "rb"))
DEEPFACE_AVAILABLE = False
try:
    os.environ["TF_CPP_MIN_LOG_LEVEL"] = "3"
    import tensorflow as tf
    from deepface import DeepFace
    DEEPFACE_AVAILABLE = True
except Exception as e:
    DEEPFACE_ERROR = str(e)
EMOTION_DEPRESSION_WEIGHTS = {
    "sad": 0.90, "fear": 0.70, "disgust": 0.50,
    "angry": 0.55, "contempt": 0.45, "neutral": 0.25,

```

```

    "surprise": 0.15, "happy": 0.05,
}
if __name__ == "__main__":
    os.makedirs("uploads", exist_ok=True)
    app.run(debug=True, port=5000)

```

Appendix C: Project Directory Structure

mindscan/

```

├── app.py          # Flask application (250 lines)
├── train.py        # Model training script (73 lines)
├── model/
│   ├── model.pkl   # Trained Logistic Regression model (~80 KB)
│   └── vectorizer.pkl # Fitted TF-IDF vectorizer (~392 KB)
├── templates/
│   └── index.html  # Frontend web interface (520 lines)
├── uploads/        # Temporary image storage (auto-cleaned)
└── env/            # Python virtual environment

```

Appendix D: API Documentation

POST /predict/text

Request:

```

json
{
  "text": "I feel so hopeless and alone..."
}

```

Response:

```

json
{
  "is_depressive": true,
  "prediction": "Depressive",
  "confidence": 94.2,
  "source": "text"
}

```

POST /predict/face

Request:

```
json
{
  "image": "data:image/jpeg;base64,/9j/4AAQ..."
}
```

Response:

```
json
{
  "is_depressive": true,
  "prediction": "Depressive",
  "confidence": 90.0,
  "dominant_emotion": "sad",
  "emotions": {
    "sad": 45.2, "fear": 15.8, "neutral": 20.1,
    "angry": 8.3, "disgust": 5.1, "surprise": 3.2, "happy": 2.3
  },
  "source": "face"
}
```

POST /predict/combined**Request:**

```
json
{
  "text": "I feel hopeless...",
  "image": "data:image/jpeg;base64,..."
}
```

Response:

```
json
{
  "text": {"is_depressive": true, "confidence": 94.2, "prediction": "Depressive"},
  "face": {"is_depressive": true, "confidence": 90.0, "prediction": "Depressive",
    "dominant_emotion": "sad", "emotions": {...}},
  "fused": {"is_depressive": true, "confidence": 92.5, "prediction": "Depressive"}
}
```

Appendix E: Installation and Setup Guide

bash

Step 1: Clone/download the project

cd mindscan

Step 2: Create virtual environment

python -m venv env

env\Scripts\activate # Windows

Step 3: Install dependencies

pip install flask pandas numpy scikit-learn nltk pillow

pip install tensorflow deepface # For face analysis

Step 4: Download NLTK data

python -c "import nltk; nltk.download('stopwords'); nltk.download('wordnet')"

Step 5: Train the model (run once)

python train.py

Step 6: Start the application

python app.py

Step 7: Open browser

Navigate to <http://localhost:5000>

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